

Standard Operating Procedure & Troubleshooting Manual

Equipment: Desiccant Dehumidifying Dryer D-301 & Drying Hopper

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Plant Location: PET Preform Line 1

Applicability: Process Operators, Maintenance Technicians, Shift Supervisors

1. Equipment Description

Dryer **D-301** dries PET pellets before they enter the extruder of preform line 1. A twin-tower desiccant unit supplies dry, hot process air to a 2,000 kg insulated drying hopper. One desiccant tower dries the air while the other is regenerated by a separate regeneration heater; the towers switch automatically every 4 hours. Design throughput is 300–400 kg/h. PET is hygroscopic: wet PET **hydrolyses** in the extruder, its intrinsic viscosity (IV) drops, and the preforms become brittle.

2. Operating Envelope & Alarm Limits

Table 1: Operating envelope and alarm limits

Parameter	Normal Range	Warning	Critical Alarm	Trip / Action
Process air dew point	–40 °C or lower	above –30 °C	above –20 °C	Quality hold on dried PET
Process air inlet temperature	160 – 170 °C	above 175 °C or below 155 °C	above 180 °C or below 150 °C	185 °C: heater cut-off
Regeneration temperature	260 – 300 °C	below 240 °C	below 220 °C	Switch to standby tower
Hopper level	60 – 90 %	below 50 %	below 40 %	Reduce extruder throughput
Residence time in hopper	4 – 6 h	below 4 h	below 3 h	Reduce extruder throughput
Pellet moisture at hopper outlet	50 ppm or less	above 50 ppm	above 100 ppm	Stop feeding extruder

Residence time = hopper material content (kg) ÷ throughput (kg/h). Pellet moisture is measured by the line technician every 2 hours.

3. Fault Matrix

Table 2: Faults, signs, causes and actions

Fault	Typical Signs	Likely Causes	Operator Action
FAULT-A: Desiccant saturation / regeneration failure	• Dew point rising above –30 °C • Pellet moisture rising • Regeneration temperature low or tower not switching • Process air temperature NORMAL	Regeneration heater failure, tower switching valve stuck, or desiccant at end of life.	1. Check regeneration heater and tower switching. 2. Switch to the standby tower manually. 3. Put PET dried since the last good test on quality hold. 4. Reduce throughput if moisture is above 50 ppm.
FAULT-B: Short residence time	• Pellet moisture rising • Dew point and air temperature NORMAL • Hopper level low or throughput above 400 kg/h	Hopper allowed to run low, throughput increased, or funnel flow (pellets channel through the centre of the	1. Restore hopper level above 60 %. 2. Reduce throughput to keep residence time at 4 h or more. 3. Check the hopper for funnel

		hopper).	flow.
FAULT-C: Low drying air temperature	• Process air inlet below 155 °C • Pellet moisture rising • Dew point NORMAL	Process heater element failure, thermocouple fault, or air flow set too high.	1. Check heater current and thermocouple. 2. Quality hold on affected PET. 3. Do not raise the setpoint above 175 °C.
FAULT-D: Over-drying / thermal yellowing	• Pellets yellow and dusty • IV lower than specification • Air temperature above 175 °C, or residence time above 8 h at low throughput	Wrong setpoint, or dryer left at full temperature during a long line stop.	1. Set dryer to standby (140 °C) during stops longer than 30 minutes. 2. Check colour and IV of hopper material. 3. Quality hold if out of specification.

4. Operating Rules

- Never feed PET with moisture above 50 ppm to the extruder.
- Keep the hopper level at or above 60 %; a low hopper shortens residence time even when the dryer is working.
- During line stops longer than 30 minutes, set the dryer to standby temperature (140 °C).
- When a moisture fault is found, quality-hold all PET dried since the last good moisture test.
- Regeneration surfaces reach 300 °C. Lock out the dryer before opening any panel.